TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Boundary: solid for ≤ or ≥, dashed for < or >.
Test (0, 0): true → shade its side; false → shade the other side.
Words: at least ≥ · at most ≤ · more than > · less than <.

1. Is (0, 0) a solution of $y > 2x - 5$?

Tip: substitute (0, 0)

A. No, because $0 < -5$

B. Yes, because $0 = 0$

C. Yes, because $0 > -5$

2. Solve: $x + y = 8$ and $x - y = 2$.

Tip: add the equations

F. (3, 5)

G. (6, 2)

H. (5, 3)

3. Which inequality means “at least 20”?

> **Tip:** at least → ≥

A. $x \geq 20$

B. $x > 20$

C. $x \leq 20$

4. Which describes the graph of $x < -2$ on a number line?

> **Tip:** < or >: open circle

F. closed circle at -2, shade left

G. open circle at -2, shade left

H. open circle at -2, shade right